

Hybrid metal-dielectric resonator for anapole-assisted enhanced chiral sensing applications

Guillermo Serrera, Javier Gonzalez-Colsa & Pablo Albella
Department of Applied Physics, University of Cantabria, Santander, Spain.

ABSTRACT

Keywords: Times Roman, image area, acronyms, references

The concept of chirality, referring to lack of specular symmetry in any plane of an object, and the resulting existence of two mirror images (or enantiomers) of the same object; is of utmost importance for the biological functionality of many molecules in the human body and pharmacological compounds. This fact has led to a growing industrial interest into the discrimination of molecular chirality. Enhancing the intrinsically weak circular dichroism (CD) chiroptical effect remains a key challenge for efficient chiral sensing and separation. The design of nanophotonic platforms for this purpose has been a hot topic for the last decade [1].

Approaches to face this challenge have been done from the fields of plasmonics, exploiting the huge field enhancement produced by the SPR phenomenon [2], or from the use of High Refractive Index Dielectric (HRID) materials due to their low-loss nature and magnetic resonances [3]. Hybrid structures, presenting both enhanced electric and magnetic resonances, have also been proposed, providing more advantageous chiroptical conditions for sensing [4].

Here, we will present our current research on this topic. Using numerical simulations aided by multipole decomposition analysis, we demonstrate how a simple hybrid gold-silicon metasurface can exploit the superposition of several resonances with non-radiating anapole states, providing remarkable conditions for the enhancement of the CD effect in the near-infrared spectrum. We will also show how the usage of these anapole states can be linked to enhanced Third Harmonic Generation, paving the way towards background-free CD signals [5,6].

REFERENCES

- [1] E. Zor, H. Bingol and M. Ersoz, Trends Analyt Chem 121(2019), 115662.
- [2] M. Hentschel, et al. Sci. Adv. 3(2017), 160273.
- [3] K. Tanaka et al. ACS Nano 14(2020), 15926.
- [4] E. Mohammadi et al. ACS Photonics 8 (2021), 1754-62.
- [5] T. Shibanuma et al. Nano Lett. 17(2017), 2647-51.
- [6] J. Gonzalez-Colsa et al. J Phys Chem Lett., 13,6230-34.